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Driving effects of U.S. monetary policy and geopolitical risks on gold reserve share

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ABSTRACT

Using quarterly data from 105 countries from 2000 to 2023 and a theoretical model of central bank reserve portfolios that incorporates gold reserve, this study investigates the effects of United States (U.S.) monetary and geopolitical risks policy on the national gold reserve share. Results show that expansionary U.S. monetary policy, which increases dollar liquidity and lowers interest rates, generally prompts central banks to raise their gold reserve shares. A stronger positive correlation between the value of the U.S. dollar and gold assets further encourages this shift. Moreover, rising geopolitical risk has emerged as a key driver of central banks to increase their gold reserve share. Findings underscore gold's expanding role as a safe haven and a reserve asset comparable to the U.S. dollar. In response to U.S. monetary policy dynamics, China should pursue more diversified reserve asset strategies, enhance gold reserve management, and continue advancing Chinese yuan internationalization.

1. Introduction

Gold, as a vital reserve asset, has long been influenced by United States (U.S.) monetary policy (Pierdzioch et al., 2014; Qian et al., 2019). Traditionally, rising U.S. interest rates have increased the opportunity cost of holding gold, thereby reducing its demand. However, recent trends reveal a noticeable weakening of this historically strong negative correlation. In 2022, despite aggressive interest rate hikes by the Federal Reserve and a sharp increase in real U.S. dollar interest rates, gold prices remained resilient and even showed an upward trend. Notably, central banks have been significantly increasing their gold reserves. According to data and surveys from the World Gold Council (WGC), global central banks collectively purchased 1136 tons of gold in 2022, marking a year-over-year increase of 152.3 %. Furthermore, in 2024, 81 % of surveyed central banks indicated that global gold holdings would continue to rise over the next 12 months. These findings suggest that central banks are actively expanding their gold holdings to increase their gold reserve share.

The surge in global gold reserves signals a major shift in how central banks manage their reserve assets. Following the 2008 global financial crisis, the U.S. implemented unconventional monetary policies, leading to depreciation of the U.S. dollar and declining returns on U.S. dollar-denominated assets. Since then, emerging economies expanded their gold reserves, while developed economies ceased gold sales, together reversing the prior trend of global central banks collectively selling gold. Between 2008 and 2023, the gold reserves of Russia, China, India, and other countries grew significantly. Notably, Russia, China, and India significantly expanded their gold reserves, increased from 519.6 to 2332.7 tons, 600 to 2235.4 tons, and 357.7 to 803.6 tons, respectively. As of November 2023, China's cumulative gold reserves had reached 71.58 million ounces, reflecting consistent month-on-month increases since early 2023, with an annual growth rate of approximately 9.92 %.

Several factors have driven this transition. First, the rapid accumulation of U.S. sovereign debt and a liquidity crisis in U.S. treasuries have raised concerns about their long-term stability (He et al., 2022; He & Krishnamurthy, 2020). Second, persistently low interest rates have eroded yields on U.S. dollar-denominated assets, diminishing their relative appeal compared to stable, non-yielding assets such as gold (Gopalakrishnan & Mohapatra, 2018a). Third, increasing financial sanctions and rising geopolitical risks have weakened confidence in the

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dollar, motivating central banks to diversify their reserve asset portfolios (McDowell, 2021; Xu & Xiong, 2022).

Against this backdrop, we address two central questions. (1) How does U.S. monetary policy influence central banks' adjustments to the gold share of their reserve portfolios? (2) To what extent does geopolitical risk, as a non-economic factor, drive central banks to increase their gold reserve share? By examining the relationship between shifts in U.S. monetary policy and central bank gold allocations, this study highlights the distinctive influence of geopolitical risk on these allocation decisions.

In recent years, the issue of the adjustment of the structure of international reserves has received increasing attention in the academic field, especially against the backdrop of increasing geopolitical tensions and the evolution of the US monetary policy stance. However, the existing literature (Ahmed & Rebucci, 2024; Chinn et al., 2024;Gopinath & Stein, 2018; Lu et al., 2024) mostly focuses on the share of the US dollar in reserves, ignoring the role of gold as a strategic reserve asset. The key to this theoretical and empirical gap is that most traditional models (Gopinath & Stein, 2018; Lu et al., 2024) set the reserve selection as a binary trade-off between the US dollar and non-US dollar currencies, ignoring the unique attributes of gold as a non-sovereign hedge against US dollar risks. In addition, although recent studies (Ngo et al., 2024; Trikounaki & Atsalakis, 2024) have begun to incorporate geopolitical factors into the analysis of reserve composition, they have not yet systematically explored how geopolitical risks and US monetary policy work together to influence the central bank's preference for gold and fiat currencies. This paper aims to fill the above research gap by constructing a portfolio allocation model that incorporates gold and traditional reserve currencies, and empirically examines the impact of the US interest rate cycle and geopolitical tensions on the central bank's gold holdings based on cross-national panel data, thereby revealing the neglected relationship mechanism between monetary policy, geopolitical spillover effects and the strategic role of gold in international reserve management.

The potential marginal contributions of this paper are reflected in three key areas. First, this study incorporates the approach of Lu et al. (2024) to extend the Gopinath and Stein (2018) model by including gold as part of the central bank's reserve portfolio. This provides a theoretical foundation for analyzing how U.S. monetary policy shapes reserve asset adjustments. Second, the paper argues that expansionary U.S. monetary policy, through increased dollar liquidity and lower interest rates, encourages central banks to raise the share of gold in their reserves. Moreover, the influence of U.S. monetary policy varies across countries, depending on their gold strategies and exchange rate systems. This perspective offers insight into the heterogeneous transmission of international monetary policy and informs national reserve management decisions. Third, this paper emphasizes that rising geopolitical risk has become a key driver prompting countries to increase their gold reserve shares. The findings aim to support central banks in optimizing reserve asset management, mitigating external shocks, and enhancing economic and financial stability in an increasingly uncertain global environment.

We organize the remainder of this paper as follows: Section 2 provides a review of the relevant literature and develops the theoretical framework. Section 3 describes the empirical model, variable selection, and definitions. Section 4 reports the empirical results on the effects of U.S. monetary policy on the gold reserve share. Finally, Section 5 concludes with a discussion of policy implications.

2. Literature review and theoretical analysis

2.1. Literature review

We contribute to three key strands of recent literature. The first examines the spillover effects of U.S. monetary policy on the structure of reserve assets. As the U.S. dollar remains the dominant currency for trade settlement, international debt financing, and reserve holdings, U.S. monetary policy exerts asymmetric spillover effects on other economies (Dahlhaus & Vasishtha, 2020; Eichengreen & Gupta, 2015; Georgiadis, 2016; Maggiori et al., 2020). Within the U.S. dollar-centric global credit system, central banks maintain substantial U.S. dollar-denominated holdings as part of their foreign exchange reserves (Ilzetzki et al., 2019; Ito & McCauley, 2020).

U.S. monetary policy affects gold and dollar reserves through liquidity and interest rate channels. When the U.S. implements expansionary policies (e.g., lowering the federal funds rate or conducting large-scale asset purchases), dollar interest rates and U.S. Treasury yields decline. This reduces the opportunity cost of holding gold, pushing its price upward (Qian et al., 2019; Zhu et al., 2018). Moreover, expansionary U.S. policy increases depreciation pressure on the dollar, weakening its role as the dominant reserve currency and prompting central banks to diversify their reserve assets as a hedge against dollar depreciation (Arslanalp et al., 2022; Gopalakrishnan & Mohapatra, 2018a).

Eichengreen et al. (2019) argued that countries' military or diplomatic ties with the U.S. reinforce their preference for dollar, thereby sustaining dollar dominance. However, emerging factors, such as the COVID-19 pandemic and global supply chain disruptions, may accelerate shifts in the global reserve asset structure (Iancu et al., 2022; Kuehnlenz et al., 2023). Although prior studies have focused on how U.S. policy affects reserve diversification, the specific drivers behind changes in the gold share of reserve assets remain relatively unexplored.

Second, a growing literature exists on gold's function as a reserve asset. Known for its high liquidity and low-risk profile, gold tends to display a strong negative correlation with equity assets during periods of financial instability. In extreme market conditions, gold acts as a safe-haven asset (Agyei-Ampomah et al., 2014; Baker et al., 2020; Baur & Smales, 2020; Bredin et al., 2015; Ciner et al., 2013; Mensi et al., 2023; Wen et al., 2022). Developed and emerging economies recognize gold as a key component of reserve asset management. Ghosh (2016), using data from 100 central banks between 1998 and 2014, found that local currency fluctuations against the dollar and rising inflation contributed to increased gold reserve shares. Gopalakrishnan and Mohapatra (2018b) further noted that when U.S. monetary policy intensifies global financial risk, central banks respond by increasing gold reserves to shield against external shocks. Maintaining large gold reserves enhances perceptions of a nation's creditworthiness and reduces sovereign risk during crises (Rathi et al., 2022). Moreover, gold reserve shares positively respond to dollar volatility. Developed economies aim to diversify their foreign exchange assets, while emerging markets focus on increasing gold reserves to rebalance their reserve portfolios (Arslanalp et al., 2023).

Third, recent literature has highlighted the impact of geopolitical risk. Events such as COVID-19, the Russia–Ukraine conflict, and escalating U.S.–China geopolitical rivalry have introduced new global uncertainties. U.S. financial sanctions and geopolitical confrontations have diminished confidence in the dollar as a reserve currency (McDowell, 2023), leading central banks to diversify into gold. Heightened geopolitical tension increases global economic uncertainty, further reinforcing gold's role as a safe haven. Gold price trends have also emerged as effective indicators of geopolitical risk, as governments increase gold holdings to hedge against potential losses (Ali et al., 2023; Chiang, 2022; Li et al., 2023; Qin et al., 2023; Su et al., 2022). Ngo et al. (2024) empirically showed that during periods of elevated geopolitical risk, countries tend to increase the ratio of gold reserves to external debt. Thus, global central banks increasingly rely on gold to maintain financial security in uncertain geopolitical environments.

Although the extant research has mainly examined U.S. monetary policy's influence on gold prices and related spillovers, it offers limited insight into how monetary policy shifts alter the gold reserve share held by central banks. Additionally, major geopolitical developments are reshaping global financial patterns, with elevated risks potentially accelerating gold accumulation. This paper investigates the impact of U.

S. monetary policy on central bank gold reserve allocations while accounting for geopolitical risks as a non-traditional economic factor. The findings aim to explain recent increases in global gold reserves and contribute to the broader literature on reserve asset adjustment. This paper provides a key analytical tool for understanding the changes in the structure of global reserve assets, such as the central bank's gold purchases reaching a historical peak in 2023, by constructing a dynamic optimization model that includes policy expectations.

2.2. Theoretical analysis

Our study is also inspired by Lu et al. (2024) who study the changes in the weights of the US dollar and the euro. Our model also differs from Gopinath and Stein (2018) who focus only on the trade-off between the US dollar and local currency assets. Our model explicitly introduces gold as a third reserve asset class. In this way, we are able to capture richer alternative dynamics under US dollar interest rate changes and geopolitical risk shocks, thereby more comprehensively analyzing the central bank's allocation behavior.

We assume the global economy comprises the U.S. and a peripheral economy, which includes three types of agents, namely, households, banks, and central banks. The model spans two periods, labeled 0 and 1. It includes three asset types: domestic currency assets, D_h; U.S. dollar-denominated assets, $D_\$$; and gold assets, D_G. At time 0, the exchange rate of the domestic currency to the dollar and gold price are normalized to 1. At time 1, the exchange rate and gold price relative to the domestic currency become stochastic. Higher values indicate depreciation of the domestic currency. Parameter $z(r_\$)$ represents the extent of domestic currency depreciation against the dollar, which is positively influenced by changes in U.S. monetary policy, $r_\$$. The probability of domestic currency depreciation is set at $p = 0.5$, with depreciation reaching $(1 + z)$.

The correlation coefficient, ρ, between dollar and gold prices ranges from -1 to 1, indicating the degree of their relationship. When ρ approaches -1, the assets are strongly negatively correlated, consistent with gold's traditional hedging role against dollar depreciation. However, as ρ increases toward positive values, this inverse relationship weakens. Under external shocks (e.g., heightened geopolitical risk) the correlation may become positive, indicating that gold, similar to the U.S. dollar, serves as a safe asset (Ding et al., 2025; Qian et al., 2019). Assuming the probability of simultaneous appreciation or depreciation of both assets is $(1 + \rho)/4$, the probability that gold appreciates during a banking crisis is $(1 + \rho)/2$.

2.2.1. Household sector

We assume that household utility is linear in consumption across time periods $T = 0$ and $T = 1$, with added utility derived from holding U.S. dollars, gold, and domestic core assets. The household utility function is given by

$$C_0 + \beta E(C_1) + \theta\alpha_\$ \log(D_\$) + \theta\alpha_G \log(D_G) + \delta(1 - \alpha_\$ - \alpha_G)\log(D_h) \tag{1}$$

where β is the time preference factor and θ and δ represent utility weights for holding U.S. dollar and gold assets, respectively. $D_\$$ is the amount of U.S. dollar-denominated assets held, D_h represents local currency assets, and $\alpha_\$$ and α_G are the proportions of assets held in dollars and gold, respectively.

Let Q_h denote the price of domestic safe assets at $T = 0$ (i.e., $Q_h = 1/(1 + r_h)$), which depends on domestic factors. Similarly, $Q_\$ = 1/(1 + r_\$)$ is the price of U.S. dollar-denominated assets, driven by global market forces such as U.S. monetary policy. Let $Q_G = 1/(1 + r_G)$ be the price of gold, where r_G is the return on gold. Maximizing household utility with respect to asset allocation gives the following first-order conditions:

$$Q_h = \beta + \delta(1 - \alpha_\$ - \alpha_G)$$

$$Q_G = \beta + \theta\alpha_G / D_G$$

$$Q_\$ = \beta + \theta\alpha_\$ / D_\$$$

Q_h is determined domestically, while $\alpha_\$$, α_G, $Q_\$$, and Q_G are governed by international markets and are therefore exogenous to peripheral economies. Due to the "exorbitant privilege" of the dollar, dollar returns are assumed to be the lowest, implying $Q_\$ > Q_G$. Let $S_\$$ and S_G denote the spreads between foreign and domestic assets:

$$S_\$ = \frac{Q_\$}{Q_h} - 1 = \frac{Q_\$}{\beta + \delta(1 - \alpha_\$ - \alpha_G)} - 1$$

$$S_G = \frac{Q_G}{Q_h} - 1 = \frac{Q_G}{\beta + \delta(1 - \alpha_\$ - \alpha_G)} - 1$$

2.2.2. Banking sector

The banking sector comprises commercial banks, financial institutions, and domestic firms. Banks borrow from households and invest in N projects. Under normal conditions, project returns are γN with $\gamma > 1$, ensuring profitability. Projects are denominated in domestic currency, but banks may borrow in foreign currencies, exposing them to exchange rate risk. In a crisis, each bank faces an independent probability m of default, resulting in zero returns. The bank's expected income is $(1 - pqm)\gamma N$.

Bank liabilities include domestic liabilities B_h backed by domestic assets, dollar liabilities $B_\$$ backed by U.S. dollar assets, and gold liabilities B_G backed by gold. These are subject to the following asset-liability constraint:

$$N = Q_h B_h + Q_\$ B_\$ + Q_G B_G \tag{2}$$

However, banks may incur the five cost types outlined in Table 1.

Banks maximize their expected profits as follows:

$$max_{B_h, B_\$, B_G} (1 - pqm)\gamma(Q_h B_h + Q_\$ B_\$ + Q_G B_G) - (C_1 + C_2 + C_3 + C_4 + pC_5) \tag{3}$$

Solving the first-order conditions for Ea. (3) yields the optimal sizes of dollar and gold liabilities:

$$B_\$ = (1/\phi p)[(1 - pqm)S_\$ + pqmz] \tag{4}$$

$$B_G = (1/\phi p)[(1 - pqm)S_G + \rho pqmz] \tag{5}$$

2.2.3. Central bank

We assume that the central bank's primary objective is to protect depositors during a banking crisis while minimizing tax burdens. To achieve this, it can take two actions: first, hold a certain share of reserve assets before the crisis (i.e., at $T = 0$); second, after the crisis (i.e., at $T = 1$), the government imposes distortionary taxes on the household sector to finance bank rescues.

Since the model includes two types of reserve assets, the central bank must optimize its allocation strategy to minimize overall holding costs

Table 1
Expected costs of bank repayment under various conditions.

Conditions	Expected costs
The depreciation of both the U.S. dollar and gold	$C_1 = [(1+\rho)/4][B_h + (1-z)(B_\$ + B_G)]$
The depreciation of the U.S. dollar and the appreciation of gold	$C_2 = [(1-\rho)/4][B_h + (1-z)B_\$ + (1+z)B_G)]$
The depreciation of gold and the appreciation of the U. S. dollar	$C_3 = (1-a)][p(1-q) + pm(1-m)][B_h + (1+z)B_\$ + (1-z)B_G)$
The appreciation of both the U.S. dollar and gold	$C_4 = a[p(1-q) + pm(1-m)][B_h + (1-z)(B_\$ + B_G)]$
Costs of currency mismatch	$C_5 = \phi/2 \times (B_\$^2 + B_G^2)$

Notes: This table illustrates the costs that the bank may encounter under various conditions where the bank does not fail.

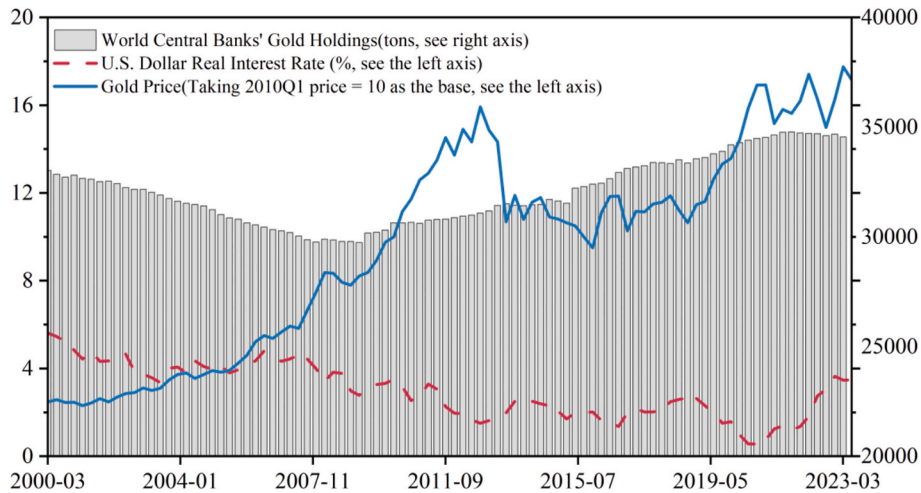

Fig. 1. Changes to the real U.S. dollar interest rate, amount of gold held by central banks worldwide, and price of gold from 2000 to 2023. Source: WGC, FRED. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

while fulfilling its stabilization objective. In times of crisis, U.S. dollar-denominated foreign exchange reserves tend to appreciate relative to the domestic currency and typically offer more effective market intervention capabilities than gold. However, gold may appreciate or depreciate in the short-term depending on broader economic conditions. Furthermore, during periods of monetary easing, the return on dollar reserves is generally lower than that of gold, especially when the U.S. follows an expansionary policy. Therefore, the central bank's balance sheet constraint at time $T = 0$ is as follows:

$$Q_{\$}R_{\$} + Q_G R_G = Q_h B_h^c \tag{6}$$

B_h^c represents interest-bearing domestic currency bonds issued by the central bank (e.g., central bank bills). The interest-bearing local currency bonds B_h^c issued by the central bank are held by banks, and $Q_h B_h^c$ appears symmetrically on both the asset and liability sides of the bank's balance sheet, offsetting each other. Therefore, the role of $Q_h B_h^c$ is not reflected in eq. (2). At $T = 1$, the central bank uses its dollar and gold reserves, along with tax revenues, to redeem maturing domestic currency bonds. The opportunity cost of holding these reserves is $C = S_{\$}R_{\$} + S_G R_G$. In the event of a crisis (with probability pq), the central bank must raise funds to support failing banks. Two scenarios arise: (1) if gold appreciates (with probability $a = (1+\rho)/2$), the central bank imposes tax T_1 on the household sector; (2) if gold depreciates (with probability $1 - a$), the central bank imposes tax T_2.

$$C + 0.5\left[pqa\gamma {T_1}^2 + pq(1-a)\gamma {T_2}^2\right] \tag{7}$$

Therefore, under the first-order condition for minimizing the central bank's total cost, the optimal sizes of foreign exchange reserves, gold reserve assets, and the taxes T_1 and T_2 imposed on the household sector are as follows:

$$R_{\$} = \frac{1}{2z}\left(2m\left[(z - S_{\$})B_{\$} - S_G B_G + \frac{N}{Q_h}\right] - T_1 - T_2\right)$$

$$R_G = \frac{1}{2z}(2mzB_G - T_1 + T_2)$$

$$T_1 = \frac{S_{\$} + S_G}{2pqa\gamma z}; T_2 = \frac{S_{\$} - S_G}{2pq(1-a)\gamma z} \tag{8}$$

Returning to the model's central aim (i.e., evaluating how changes in U.S. monetary policy affect the gold reserve share in total reserve assets) we note that $ShareR_G = R_G/(R_G + R_{\$})$ is an increasing function of the ratio $R_G/R_{\$}$. Taking the derivative of this ratio with respect to the U.S. interest rate yields the following.

$$R_G \Big/ R_{\$} = \frac{2mzB_G - T_1 + T_2}{2m\left[(z - S_{\$})B_{\$} - S_G B_G + \frac{N}{Q_h}\right] - T_1 - T_2} \tag{9}$$

$$\begin{aligned}\frac{\partial(R_G/R_{\$})}{\partial(r_{\$})} &= \frac{\partial(R_G/R_{\$})}{\partial(B_G/B_{\$})}\frac{\partial(B_G/B_{\$})}{\partial(z)}\frac{\partial(z)}{\partial(r_{\$})} \\ &= \frac{\partial(R_G/R_{\$})}{\partial(B_G/B_{\$})}\frac{pqm}{{B_{\$}}^2}\left[(1-pqm)(\rho S_{\$} - S_G)\right]\frac{\partial(z)}{\partial(r_{\$})}\end{aligned} \tag{10}$$

The term $\partial(R_G/R_{\$})/\partial(B_G/B_{\$})$ reflects the central bank's synchronized reserve strategy in response to the liabilities of other sectors and is generally positive (Gopinath & Stein, 2018). Prior research and historical evidence suggest that gold prices typically exhibit a negative correlation with U.S. interest rates, implying that the value of Eq. (10) is negative. During periods of expansionary U.S. monetary policy, as the Federal Reserve increases the money supply and purchases bonds, market interest rates decline, and the dollar weakens. In response, central banks reevaluate their reserve portfolios in line with the three core principles of reserve management (i.e., safety, liquidity, and profitability). Moreover, to counter reduced returns on U.S. dollar-denominated assets, central banks often increase gold holdings, using gold as a hedge against dollar depreciation.

However, if the correlation ρ between dollar and gold asset values becomes positive, Eq. (10) may turn positive. In such cases, even if the U.S. implements monetary tightening—characterized by rising interest rates and an appreciating dollar—gold prices may increase alongside the dollar due to heightened geopolitical risks. For example, since 2015, the negative correlation between gold and the dollar has steadily weakened. This trend was particularly notable from 2022 to 2024, when the dollar and gold reached historic highs. Furthermore, countries concerned about overdependence on dollar assets may respond by increasing their gold reserve share, prioritizing security and diversification.

Thus, this paper proposes the following research hypotheses:

H1: The expansionary U.S. monetary policy lowers dollar interest rates, prompting countries to increase their gold reserve share.

H2: As the value correlation between the U.S. dollar and gold assets intensifies, U.S. tightening monetary policy encourages countries to boost their gold reserve share.

3. Methodology and data

3.1. Benchmark regression model

According to our extended research framework, the empirical model

should capture the dynamic response of gold reserves to changes in US monetary policy. To reduce omitted variable bias, we selected U.S. monetary policy as the core explanatory variable and introduced gold prices, global financial market conditions, and the macroeconomic conditions of various countries as controls. This approach reduces the likelihood of excluding important explanatory factors. The regression model is specified as follows:

$$Goldr_{i,t} = \alpha_0 + \alpha_1 USMP_{t-1} + \sum_{k=1}^{n} \beta_k GFM_{t-1} + \sum_{k=2}^{n} \alpha_k MacE_{i,t-1} + \mu_i + \varepsilon_{i,t} \tag{11}$$

where $Goldr_{i,t}$ represents the gold reserve share held by country i at time t relative to total reserve assets; $USMP_t$ denotes U.S. monetary policy changes; GFM_t reflects global financial market conditions; $MacE_{i,t}$ captures country-level macroeconomic factors; μ_i represents country-specific fixed effects; and $\varepsilon_{i,t}$ is the random error term.

3.2. Variable definitions

We used panel data from 105 countries, excluding the U.S. to avoid endogeneity. The dataset spanned 2000–2023, covering key macroeconomic episodes, including the 2007–2008 global financial crisis, the subsequent era of U.S. quantitative easing, and COVID-19. This broad time span enables an assessment of how U.S. monetary policy influences gold reserve shares under varied macroeconomic conditions. Data were sourced from International Monetary Fund Statistics, the World Bank, and the China Economic Information Center database. The U.S. monetary policy variable is drawn from the Federal Reserve Economic Data. The capital account openness data are based on the latest dataset released by Chinn and Ito (2006). Detailed definitions are provided in Table 2.

Table 2
Variable definitions.

Type	Symbol	Definition	
Dependent	*Goldr*	Gold reserve share	Proportion of central bank gold reserves in total reserves
Independent	*USMP*	U.S. monetary policy	Quarterly average of effective federal funds rate Wu–Xia shadow federal funds rate
Control	*Price*	Gold price	Price of 1 oz. gold (logarithmic)
	VIX	Global financial risk	CBOE market volatility index (logarithmic)
	Cov	Correlation between U.S. dollar and gold prices	Correlation coefficient between gold price and U.S. dollar index in 24-month rolling window
	USBR	U.S. Treasury yield	U.S. 10-year Treasury yield
	USBS	U.S. Treasury security	Convenience yield on U.S. 1-year Treasury bond
	EXPR	Export accumulation	Ratio of total goods and services exports to GDP
	IMPR	Import consumption	Ratio of total goods and services imports to GDP
	FAR	Financial account channel	Ratio of financial account surplus to GDP
	E	Exchange rate	Exchange rate of local currency against U.S. dollar (logarithmic)
	Debtr	Foreign debt burden	Ratio of foreign debt to GDP
	Fxr	Foreign exchange reserves share	Ratio of foreign exchange reserves to GDP
	GDP	GDP growth rate	Quarter-on-quarter growth of GDP
	Kaopen	Capital account openness	Capital account openness level

Notes: This table illustrates all the variables used in the text, along with their corresponding variable types, variable symbols, and the data sources or calculation methods for each of these variables.

Table 3
Descriptive statistics.

Variable	Observations	Mean	Standard Deviation	Min	Max
Goldr	8258	0.15	0.22	0	0.995
USMP	10,080	1.79	1.94	0.06	6.51
Price	10,080	6.82	0.64	5.57	7.59
VIX	10,080	20.05	7.58	10.31	58.75
Cov	10,080	−0.50	0.32	−0.97	0.17
USBR	10,080	3.25	1.30	0.65	6.47
USBS	9870	−0.13	0.17	−0.64	0.15
EXPR	7566	0.01	0.06	−0.70	0.89
IMPR	7574	0.48	0.35	0.04	2.42
FAR	7443	−0.01	0.44	−8.33	18.33
E	9927	−2.55	2.85	−10.64	14.69
Debtr	9843	1.59	4.96	0.00	71.86
Fxr	8161	0.73	0.81	0.00	5.66
GDP	8284	0.02	0.08	−0.63	4.55
Kaopen	8924	0.61	0.37	0	1

Notes: This table illustrates the descriptive statistics for the key variables analyzed.

3.2.1. Dependent variable

The gold reserve share is defined as the proportion of a central bank's gold reserves relative to its total reserve assets. Total reserve assets include both foreign exchange and gold reserves, measured in U.S. dollars. This share reflects the relative importance of gold in reserve allocation and serves as a key indicator of a central bank's reserve management strategy.

3.2.2. Independent variable

Following Kim and Nguyen (2009) and Swanson (2021), we used the effective federal funds rate (EFFR) as a proxy for U.S. monetary policy. The EFFR is the weighted average of overnight interbank lending rates, which the Federal Reserve adjusts toward its target rate through open market operations. Since the introduction of quantitative easing, the impact of U.S. monetary policy on market interest rates has intensified. To also capture the influence of unconventional policy tools, we employ the shadow rate from the Federal Reserve Bank of Atlanta, following the method of Wu and Xia (2016). The shadow rate reflects the Fed's use of unconventional tools such as quantitative easing and forward guidance when nominal interest rates approach zero. It is included as a robustness check.

3.2.3. Control variables

Global financial market controls include the following.

- Gold price: Measured as the natural logarithm of the gold price published by the WGC, used to assess the effect of gold price fluctuations
- Global financial risk: Captured using the VIX index, which represents investor sentiment and is a key indicator of the global financial cycle (Miranda-Agrippino & Rey, 2020; Smales, 2022)
- Correlation between gold and the U.S. dollar: Calculated as the 24-month rolling correlation between gold prices and the U.S. dollar index. A strong negative correlation is generally expected; however, when both serve as safe assets, the correlation may turn positive, incentivizing gold accumulation
- U.S. Treasury yield: The real yield is obtained by subtracting the 10-year expected consumer price index from the 10-year U.S. Treasury bond yield. Because Treasury bonds offer interest income and gold does not, lower real yields increase gold's appeal.
- U.S. Treasury security: After the financial crisis, safe dollar assets became scarcer, leading to deviations from covered interest rate parity. This deviation (i.e., "convenience yield") reflects investors' willingness to accept lower returns in exchange for safe assets. Following Du et al. (2018), we defined this yield as follows:

$$USBS_t = -\left[i_t - i_t^* + \left(f_t^1 - s_t\right)\right] \tag{12}$$

where f_t^1 is the one-year forward exchange rate of the U.S. dollar against foreign currencies, s_t is the nominal spot exchange rate, and $i_t - i_t^*$ is the interest rate differential between U.S. and foreign one-year bonds.

Macroeconomic control variables for each country include the following.

- Trade share: The sum of exports and imports of goods and services as a percentage of gross domestic product (GDP), measuring a country's exposure to global trade.
- Financial account channel: The ratio of financial account surplus to GDP, indicating the extent to which reserve assets are accumulated through capital flows.
- Exchange rate: The nominal exchange rate of the domestic currency against the U.S. dollar, included to reflect exchange rate pressures that may influence reserve strategies.
- Foreign debt burden: Measured as the ratio of external debt to GDP. Higher external debt increases reliance on foreign exchange reserves and may reduce the gold share.
- Foreign exchange reserves share: Central banks with larger foreign exchange reserves may reallocate excess reserves from dollar assets to more diversified portfolios (Aizenman et al., 2020).
- GDP growth rate: Higher growth attracts cross-border capital inflows and may lead to increased foreign exchange reserves while lowering the relative gold share.
- Capital account openness: Measured by the Chinn–Ito index (Chinn & Ito, 2006), reflecting how freely capital can flow in and out of a country.

Descriptive statistics of the variables are presented in.

4. Empirical results

4.1. Benchmark results

We conducted a multicollinearity assessment prior to the empirical analysis. The maximum variance inflation factor was 2.72, indicating that there is no significant multicollinearity among explanatory variables. Table 4 presents the influence of changes in U.S. monetary policy on countries' gold reserve shares. Column (1) shows that the coefficient for *USMP* is −1.549, statistically significant at 5 %. The findings reveal that expansionary U.S. monetary policy significantly increases the share of gold reserves across all categories of the sample. Specifically, a 1 % decline in U.S. interest rates results in a 1.549 %, 3.032 %, and 1.379 % increase in the gold reserve share for the full sample, developed economies, and developing economies, respectively. These findings indicate that the implementation of expansionary U.S. monetary policies increases dollar liquidity and lowers interest rates, encouraging countries to raise their gold reserve share.

According to the three principles of reserve management (e.g., risk, return, and liquidity) central banks adjust their portfolios accordingly. When yields on U.S. dollar-denominated assets decline, gold becomes more attractive as a safe haven. During periods of abundant dollar liquidity and falling interest rates, central banks tend to increase their gold holdings to hedge against potential depreciation of dollar assets. Moreover, expansionary U.S. monetary policy has heightened concerns over inflation and U.S. economic prospects, prompting central banks to strengthen their gold reserves as a hedge against uncertainty. These empirical results confirm the paper's theoretical framework and support **H1**.

4.2. Robustness test

4.2.1. Replacement of the independent variable

To address potential concerns regarding variable selection, we conducted robustness tests using alternative proxies for U.S. monetary policy. First, we employ the shadow rate from the Federal Reserve Bank of Atlanta, considering unconventional monetary policy effects at the zero lower bound. Compared with Table 4, the regression coefficients for *USMP* in Table 5 differ numerically but retain the same sign and statistical significance. Second, we use the ratio of Federal Reserve assets to U.S. GDP (*FRAG*) to proxy for balance sheet expansion during quantitative easing. A higher *FRAG* indicates monetary easing and greater global liquidity (Carpenter et al., 2015). In **Error! Reference source not found.**, the coefficient for *FRAG* is significantly positive, indicating a positive relationship between Federal Reserve asset expansion and gold reserve accumulation. These results suggest that expansionary U.S. monetary policy promotes gold reserve growth as a hedge against economic and inflationary risks. Thus, the robustness of the baseline results is confirmed.

Table 4
Effects of U.S. monetary policy changes on gold reserve share of various countries.

	All	*Developed*	*Developing*
	(1)	(2)	(3)
	Goldr	*Goldr*	*Goldr*
L.USMP	−1.549**	−3.032**	−1.379*
	(0.600)	(1.273)	(0.717)
L.Price	1.517***	3.439**	1.235*
	(0.562)	(1.299)	(0.724)
L.VIX	−0.027**	−0.055**	−0.025*
	(0.011)	(0.023)	(0.013)
L.Cov	0.384***	0.891**	0.293*
	(0.124)	(0.327)	(0.163)
L.USBR	−0.137**	−0.283**	−0.118*
	(0.055)	(0.119)	(0.065)
L.USBS	−12.750**	−23.875**	−11.774*
	(5.091)	(10.665)	(5.936)
L.EXPR	−0.035	−0.034	−0.058**
	(0.027)	(0.057)	(0.022)
L.IMPR	−0.039	−0.047	−0.116
	(0.090)	(0.143)	(0.104)
L.FAR	0.002	0.000	0.005
	(0.005)	(0.006)	(0.005)
L.E	−0.027	0.140	−0.056*
	(0.024)	(0.108)	(0.028)
L.Debtr	0.002	0.002	0.008
	(0.002)	(0.002)	(0.005)
L.Fxr	−0.107***	−0.119***	−0.061**
	(0.023)	(0.025)	(0.026)
L.GDP	−0.132**	−0.433**	−0.016
	(0.053)	(0.181)	(0.030)
L.Kaopen	−0.001	−0.040	0.003
	(0.050)	(0.114)	(0.027)
Constant	12.079***	27.396**	9.658*
	(4.413)	(10.223)	(5.671)
Country FE	YES	YES	YES
Adj.R^2	0.237	0.335	0.239
Observations	5532	2405	3127

Notes: This table reports the regressions on the effects of U.S. monetary policy changes on the gold reserve share of various countries. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. *All* in the top row represents complete samples, *Developed* represents developed countries, and *Developing* represents developing countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

4.2.2. Replacement of dependent variable

To further validate the findings, we replace the gold-to-reserves ratio with the ratio of gold reserves to GDP (*Goldgdp*). As shown in Table 6, the shift to an economic-scale-adjusted measure produces numerically different coefficients for *USMP*, but the negative sign and statistical significance are preserved. This implies that Federal Reserve easing reduces the yield of traditional reserve assets like U.S. Treasury bonds, although making gold more attractive. The results confirm the

Table 5
Independent variable replacement robustness test results.

	All	*Developed*	*Developing*	*All*	*Developed*	*Developing*
	(1)	(2)	(3)	(4)	(5)	(6)
	Goldr	*Goldr*	*Goldr*	*Goldr*	*Goldr*	*Goldr*
L.USMP	−0.137**	−0.269**	−0.122*			
	(0.053)	(0.113)	(0.064)			
L.FRAG				0.016**	0.032**	0.015*
				(0.006)	(0.014)	(0.008)
Control variables	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES
Adj.R^2	0.237	0.335	0.239	0.223	0.308	0.215
Observations	5532	2405	3127	5532	2405	3127

Notes: This table reports the regressions on the effects of U.S. monetary policy changes on the gold reserve share of various countries. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. *All* in the top row represents complete samples, *Developed* represents developed countries, and *Developing* represents developing countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

Table 6
Dependent variable replacement robustness test results.

	All	*Developed*	*Developing*
	(1)	(2)	(3)
	Goldgdp	*Goldgdp*	*Goldgdp*
L.USMP	−0.850***	−1.195***	−0.874**
	(0.269)	(0.301)	(0.428)
Control variables	YES	YES	YES
Country FE	YES	YES	YES
Adj.R^2	0.290	0.385	0.291
Observations	5532	2405	3127

Notes: This table reports the regressions on the effects of U.S. monetary policy changes on the gold reserve share of various countries. *Goldgdp* represents the proportion of the central bank's gold reserves relative to its GDP. *All* in the title represent complete samples, *Developed* represents developed countries, and *Developing* represents developing countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

robustness of the regression outcomes.

4.2.3. Excluding the global financial crisis period

To eliminate bias from the extreme safe-haven demand during the 2007Q3–2009Q2 global financial crisis, we exclude that period from the sample. As shown in Table 7, the coefficient for U.S. monetary policy remains significantly negative, even with the crisis years removed. This demonstrates that central banks' accumulation of gold is not limited to crisis-driven behavior but reflects broader strategic responses to U.S. policy cycles. Thus, excluding the crisis period further strengthens the robustness of our results.

Table 7
Global financial crisis period exclusion robustness test results.

	All	*Developed*	*Developing*
	(1)	(2)	(3)
	Goldr	*Goldr*	*Goldr*
L.USMP	−0.882***	−1.216***	−0.918**
	(0.273)	(0.305)	(0.435)
Control variables	YES	YES	YES
Country FE	YES	YES	YES
Adj.R^2	0.288	0.385	0.288
Observations	5216	2260	2956

Notes: This table reports the regressions on the effects of U.S. monetary policy changes on the gold reserve share of various countries, excluding the global financial crisis period. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. *All* in the top row represent complete samples, *Developed* represents developed countries, and *Developing* represents developing countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

4.3. Heterogeneity analysis

4.3.1. Heterogeneity across different periods

Prior to the 2008 financial crisis, developed countries consistently reduced their gold reserves, contributing to the global decline in central bank gold holdings. Between 2000Q1 and the global financial crisis, developed countries' gold reserves fell by around 4000 tons from a peak of 25,056.72 tons. Meanwhile, gold holdings in developing countries remained relatively stable. After the crisis, developed economies largely ceased selling gold. We divided the sample into four monetary policy phases, namely, pre-crisis period (before 2008), U.S. unconventional policy period (2008–2014), stable period (2014–2019), and post-COVID-19 period (2020 onward), allowing for analysis of how U.S. monetary shifts affected reserve strategies across different macroeconomic regimes.

Table 8 shows that from 2000Q1 to 2008Q2, the regression coefficient of U.S. monetary policy on gold reserves was −0.090. This negative relationship suggests that as U.S. monetary policy tightened, the amount of gold in countries' reserve assets went down. As a result, countries reduced their gold holdings and shifted more of their reserves to U.S. dollar assets. This is consistent with traditional theory, where higher U.S. dollar interest rates make the opportunity cost of holding non-yielding gold higher, lowering the share of gold in central banks' reserve portfolios.

From 2008Q3 to 2014Q4, the coefficient for U.S. monetary policy remained negative but was not statistically significant. Although increased dollar liquidity boosted demand for foreign exchange reserves, inflation and uncertainty led some central banks to maintain or expand gold holdings, yielding an insignificant net change in the gold share of reserves.

Between 2020Q1 and 2022Q1, global economic shocks and ultra-low interest rates drove a significant increase in gold reserves. As central banks reduced U.S. Treasury bond holdings, gold gained importance as a safe haven and an alternative vehicle for capital allocation.

Notably, from 2015Q1 to 2019Q4, the *USMP* coefficient was positive and statistically significant. This indicates that, despite rising U.S. interest rates, countries continued to increase their gold reserve shares. As the U.S. exited quantitative easing and entered a tightening phase, foreign debt growth slowed and central banks' enthusiasm for holding foreign exchange reserves waned. Simultaneously, geopolitical tensions and the strengthening correlation between gold and dollar assets encouraged greater gold accumulation. These findings confirm that gold functions as a hedge against dollar depreciation and a strategically valuable reserve asset. During global shocks, central banks with diversified reserve portfolios are better equipped to manage market risks (Aizenman et al., 2020). The results support the theoretical foundation of this paper and validate **H2**.

Table 8
Heterogeneous effects of different periods.

	2000Q1–2008Q2	*2008Q3–2014Q4*	*2015Q1–2019Q4*	*2020Q1–2022Q1*
	(1)	(2)	(3)	(4)
	Goldr	*Goldr*	*Goldr*	*Goldr*
L.USMP	−0.090**	−0.120	0.030**	−1.383***
	(0.034)	(0.106)	(0.012)	(0.309)
Control variables	YES	YES	YES	YES
Country FE	YES	YES	YES	YES
Adj.R^2	0.244	0.142	0.085	0.132
Observations	1734	1741	1456	601

Notes: This table reports the heterogeneity analysis of U.S. monetary policy changes on the gold reserve share of various countries at different times. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

4.3.2. Heterogeneity in central bank gold reserve adjustments

Based on changes in gold holdings, countries are categorized into three groups: Surge Group (gold holdings increased by more than $1 billion), Stable Group (increases below $1 billion), and Reduction Group (countries that reduced holdings). Russia and China are prominent in the Surge Group, each increasing their gold reserves by more than 1000 tons.

According to Table 9, the tightening of U.S. monetary policy significantly reduces the gold reserve share among countries in the Surge Group. Higher interest rates raise the opportunity cost of holding gold, leading to reduced gold shares to avoid loss of yield. Conversely, during monetary easing, declining real interest rates make U.S. dollar-denominated assets less attractive, prompting shifts toward safer alternatives like gold. Gold's liquidity, inflation protection, and safe-haven qualities make it an ideal reserve asset in such conditions.

In contrast, the *USMP* coefficient is not statistically significant in the Stable and Reduction Groups. These results suggest that such countries have relatively consistent reserve strategies and are less sensitive to U.S. monetary policy cycles.

4.3.3. Heterogeneity in the exchange rate system

Using the exchange rate classification from Ilzetzki et al. (2019), countries are grouped into three categories, namely, fixed, managed floating, and free-floating exchange rate systems. According to the coarse classification data set, countries with scores of 1–2 are classified as fixed exchange rate system, countries with scores of 3 are classified as managed floating exchange rate system, and countries with scores of 4–5 are classified as free-floating exchange rate system.

Table 10 presents the regression results. In fixed exchange rate system, where the domestic currency is pegged to the U.S. dollar (or euro), central bank interventions typically occur via foreign exchange markets. In these cases, expansionary U.S. monetary policy makes gold more attractive for hedging against dollar depreciation, thereby leading to increased gold reserves.

Under the managed floating exchange rate system, although exchange rates are primarily determined by market forces, central banks still intervene using reserve assets to maintain the rate within a predetermined range. As U.S. dollar liquidity rise and interest rates decline, the demand for gold as a hedging asset increases. The central banks operating under a managed floating exchange rate system also expand their gold reserves to mitigate risks associated with potential depreciation of the U.S. dollar.

Countries that adopt a free-floating exchange rate system—typically developed economies—have historically maintained a high proportion of gold in their reserves. These nations generally refrain from intervening in foreign exchange markets and emphasize comprehensive reserve asset management less compared to those with fixed exchange rate systems (Arslanalp et al., 2023).

4.4. Geopolitical risk effects

Rising geopolitical risk has become a key driver of the recent increase in central bank gold holdings worldwide. Gold's safe-haven characteristics are particularly pronounced amid escalating geopolitical uncertainty. As global political tensions rise and deglobalization intensifies, central banks are reevaluating the safety and profitability of reserve assets, increasingly favoring gold as a "supra-sovereign" asset. To examine this relationship, we construct the following econometric model to estimate the impact of geopolitical risk on the gold reserves share.

Table 9
Heterogeneous effects of different central bank gold reserve adjustments.

	Surge	*Stable*	*Reduction*
	(1)	(2)	(3)
	Goldr	*Goldr*	*Goldr*
L.USMP	−2.235***	−1.641	1.089
	(0.714)	(1.650)	(0.737)
Control Variables	YES	YES	YES
Country FE	YES	YES	YES
Adj.R^2	0.350	0.290	0.442
Observations	3354	1219	959

Notes: This table reports the heterogeneity analysis of U.S. monetary policy changes on the gold reserve share of different central bank gold reserve adjustments. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. *Surge* represents countries whose gold holdings increased by more than $1 billion, *Stable* represents countries whose gold holdings increased by less than $1 billion, and *Reduction* represents countries that reduced their holdings. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

Table 10
Heterogeneous effects of different exchange rate systems.

	Fixed	*Managed floating*	*Free-floating*
	(1)	(2)	(3)
	Goldr	*Goldr*	*Goldr*
L.USMP	−1.337**	−3.852***	0.377
	(0.636)	(1.336)	(0.699)
Control Variables	YES	YES	YES
Country FE	YES	YES	YES
Adj.R^2	0.318	0.307	0.811
Observations	3596	1196	496

Notes: This table reports the heterogeneity analysis of U.S. monetary policy changes on the gold reserve share of various countries in different exchange rate systems. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. *Fixed* represents countries under fixed exchange rate system, *Managed floating* represents countries under managed floating exchange rate system, and *Free-floating* represents countries under free-floating exchange rate system * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

$$Goldr_{i,t} = \alpha_0 + \alpha_1 USMP_{t-1} + \alpha_2 countryrisk_{i,t-1} + \sum_{k=1}^{n} \beta_k GFM_{t-1} + \sum_{k=3}^{n} \alpha_k control_{i,t-1} + \mu_i + \varepsilon_{i,t} \quad (13)$$

where $countryrisk_{i,t}$ represents the geopolitical risk index level of country i at time t. This index is constructed using the methodology of Caldara and Iacoviello (2022), which quantifies the frequency of articles related to geopolitical risk published in major global news outlets as a proxy for geopolitical tension. Based on the quantile distribution of the index, we divide the sample into two groups: countries with a geopolitical risk index below the 25th percentile is classified as the "low geopolitical conflict" group, whereas those above the 75th percentile form the "high geopolitical conflict" group.

Table 11 reveals three key findings.

- For the full sample, tightening of U.S. monetary policy significantly negatively affects the gold reserve share. Meanwhile, geopolitical risks significantly positively affect it, with a coefficient of 0.054 at 5 %. This indicates that higher geopolitical risk is associated with a larger share of gold reserves held by central banks.
- For countries experiencing high geopolitical risks, its impacts on gold reserves remains significantly positive, with a coefficient of 0.039 at 5 %. Although the impact of U.S. monetary tightening is negative, it is not statistically significant. These findings suggest that in high-risk environments, geopolitical risks play a more dominant role in influencing central bank reserve decisions.
- In countries with low geopolitical risk, U.S. monetary tightening exerts a significantly negative effect on the gold reserve share (coefficient = −1.492 at 10 %), while geopolitical risk is not a significant factor. In these relatively stable environments, central banks appear to adjust gold reserves primarily in response to U.S. monetary policy. Overall, central banks prioritize gold accumulation in response to rising geopolitical uncertainty, while the influence of U.S. monetary policy is more pronounced in geopolitically stable environments. These results align with theoretical expectations that gold functions as a hedge against geopolitical risks (Ali et al., 2023; Ngo et al., 2024).

5. Conclusions

From an academic perspective, this paper contributes to the existing literature in three important ways. First, it proposes a novel theoretical framework that incorporates gold into central bank's reserve allocation decisions. This framework not only enriches the theoretical foundations of reserve management but also offers new perspectives for rethinking the international financial architecture in the context of an emerging multipolar monetary system, challenging the long-standing paradigm centered on U.S. dollar-denominated reserves. Second, it provides robust empirical evidence on how U.S. monetary policy influences the share of gold in countries' official reserves. Drawing on a comprehensive panel dataset of cross-national countries with quarterly observations from 2000 to 2023, we conduct an in-depth analysis of the effects of U.S. monetary policy and geopolitical risk on gold reserve allocations. The findings confirm that expansionary U.S. monetary policy, by lowering dollar interest rates, encourages countries to increase their gold reserve shares. A stronger correlation between international reserves. Drawing on a quarterly panel dataset of 105 countries spanning the period from 2000 to 2023, we conduct an in-depth analysis of the impact of U.S. monetary policy and geopolitical risks on countries' gold reserve shares. The results show that expansionary U.S. monetary policy, by lowering dollar interest rates, significantly encourages countries to increase the share of gold in their reserves. Moreover, a stronger correlation between the value of the U.S. dollar and gold assets further reinforces this reallocation. Third, the paper highlights the growing role of geopolitical risk as a key driver of gold accumulation. As global political and economic uncertainties intensify, central banks increasingly turn to gold as both a hedge against dollar depreciation and a "super-sovereign" reserve asset—one that operates independently of any single national economy. Together, these findings shed new light on the evolving strategic considerations behind reserve composition and underscore the rising prominence of gold in a shifting global monetary landscape.

Based on these empirical insights, we offer three policy recommendations:

- Diversify reserve assets

Developing countries often face vulnerabilities due to heavy reliance on U.S. dollar assets. Fluctuations in U.S. monetary policy can significantly affect the value of their reserves. With the world's largest foreign exchange reserves, China should adopt a strategy for gradually optimizing its assets while maintaining sufficient dollar liquidity. In the short term, China cannot drastically reduce its holdings of U.S. assets, as few alternatives offer the same level of safety. Therefore, U.S. dollar assets should remain at no less than 50 % of the reserve portfolio. Furthermore, a rebalancing strategy can involve gradually replacing U.S. long-term Treasury bonds with U.S. corporate bonds and blue-chip equities rated Aa2 or higher by Moody's (or AA and above by S&P), thereby reducing political risk through market-based asset replacement. In the medium term, China should implement a plan to rebuild its foreign exchange reserves; orderly increase the proportion of assets such as the euro, Singapore dollar, and Japanese yen; ensure the safety margin for international payments; and systematically reduce the risks associated with single-currency exposure.

- Strengthen gold reserve management

As central banks globally increase their gold holdings amid the current U.S. interest rate cut cycle, the opportunity for China to expand its own gold reserves is timely. Rising geopolitical tensions have reinforced gold's role as an effective hedge. Currently, Chinese gold reserves remain below those of the U.S. and leading European countries. To close this gap, China should consider gradually reducing long-term U.S. Treasury purchases and allocating a portion of new foreign exchange inflows toward gold accumulation.

- Promote the internationalization of the Chinese yuan (RMB)

The limited international role of the RMB presents structural challenges in reserve asset management, particularly given China's continued dependence on the U.S. dollar for trade and investment.

Table 11
Geopolitical conflict effects.

	All	*High*	*Low*
	(1)	(2)	(3)
	Goldr	*Goldr*	*Goldr*
L.USMP	−0.890**	−0.283	−1.492*
	(0.416)	(1.841)	(0.848)
L.countryrisk	0.054**	0.039**	0.175
	(0.022)	(0.016)	(0.410)
Control Variables	YES	YES	YES
Country FE	YES	YES	YES
Adj.R^2	0.348	0.560	0.383
Observations	3076	705	801

Notes: This table reports the regressions on the effects of U.S. monetary policy changes on the gold reserve share of various countries after incorporating geopolitical risks. The "*High*" group represents countries with a geopolitical risk index above the 75th percentile. The "*Low*" group represents countries with a geopolitical risk index below the 25th percentile. *Goldr* represents the proportion of the central bank's gold reserves relative to its total reserve assets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the country level and reported in parentheses.

However, the global momentum toward "de-dollarization" offers a strategic window for the RMB. Hence, to seize this opportunity, China should prioritize policies that encourage greater international use of the RMB by enhancing cross-border settlement infrastructure, offering tax incentives, and providing financial support to bolster the RMB's global appeal.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

Partial data of this study are available from the corresponding author upon reasonable.

AI-assisted writing disclosure.

The authors employed generative artificial intelligence and AI-assisted technologies (e.g., ChatGPT) to polish some of the content of this manuscript. All text generated by these tools was reviewed, edited, and approved by the authors, who bear full responsibility for the integrity and accuracy of the final content.

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